

GEOMETRY OF AN INVESTMENT PORTFOLIO

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- ✓ Is my own and personal opinion
- ✓ Does not represent my employer's point of view
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A bit about myself



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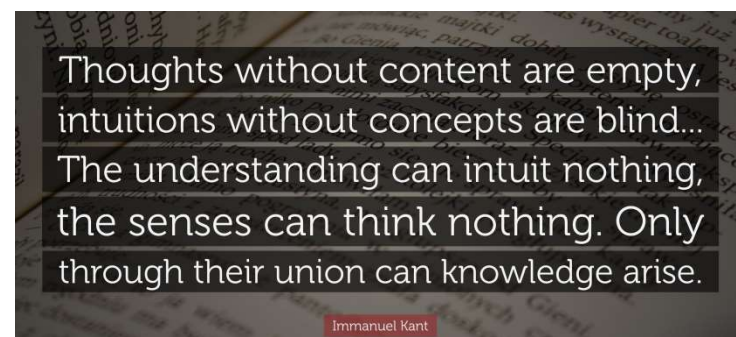
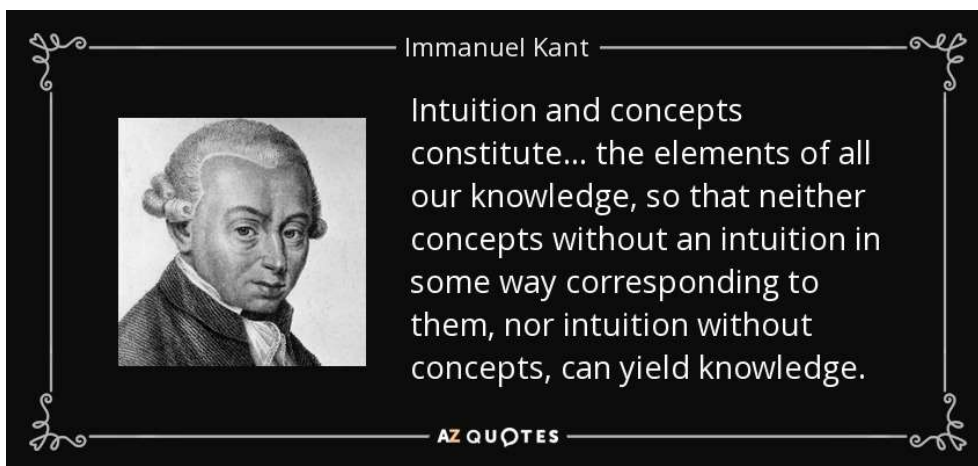
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On intuition and concepts

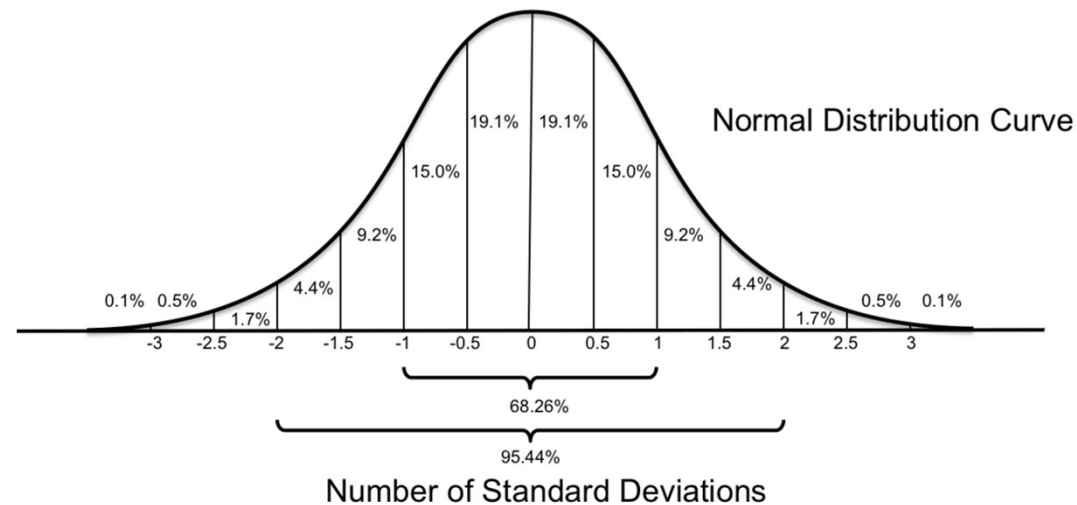
- Our goal is to present the theory of portfolio investments
 - Based on geometric intuitions and concepts
- The first thing that one should aim for understand something is intuition
 - Without intuition we cannot have a clear view of what to prove or test
- However, intuition alone is not sufficient for fully understanding
 - Concepts, theory and models are necessary



1. Geometry of confidence intervals

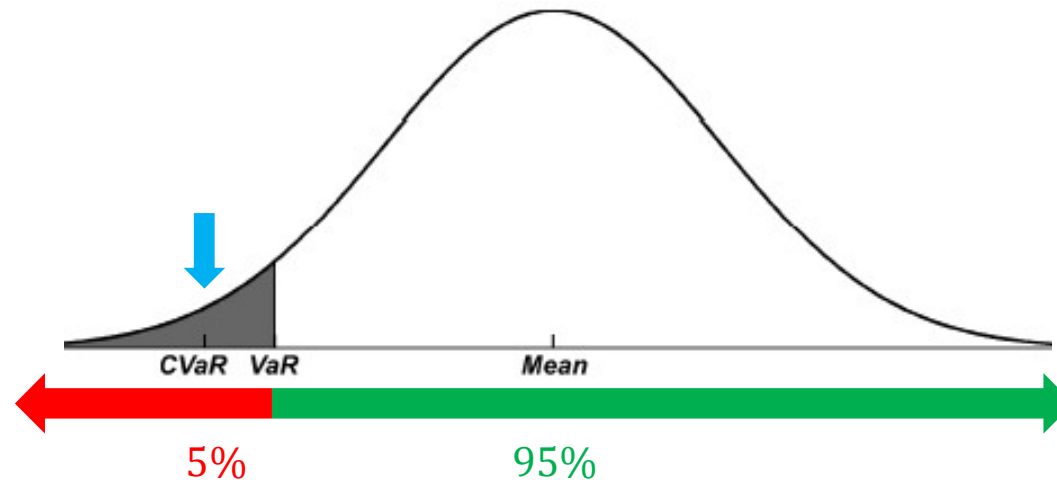
Two-sided confidence intervals

- We start with a random variable X with mean μ
- The confidence interval with level $q \in (0,1)$ around μ is
 - $CI(q) = [\mu - K(q), \mu + K(q)]$
 - Where $K(q) > 0$ is such that $P[x \in CI(q)] = q$
- For normal random variables $N(\mu, \sigma^2)$ we have
 - $CI(95\%) = [\mu - 1.96 * \sigma, \mu + 1.96 * \sigma]$



Value at Risk

- The Value at Risk of level $q \in (0,1)$ is a number $VaR(q)$ such that
 - $P[x \geq VaR(q)] = q$
- Equivalently, $P[x \leq VaR(q)] = 1 - q$
- For normal random variables $N(\mu, \sigma^2)$ we have
 - $VaR(95\%) = \mu - 1.64 * \sigma$
- Conditional Value at Risk
 - $CVaR(95\%)$ is the average of all losses at the left tail of the $VaR(95\%)$

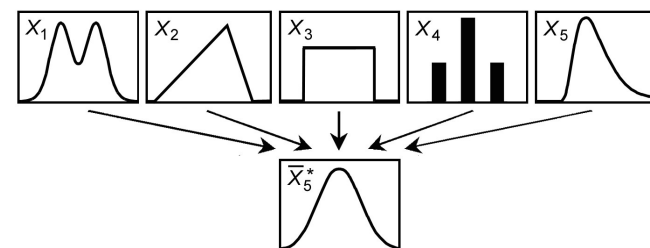


2 important theorems

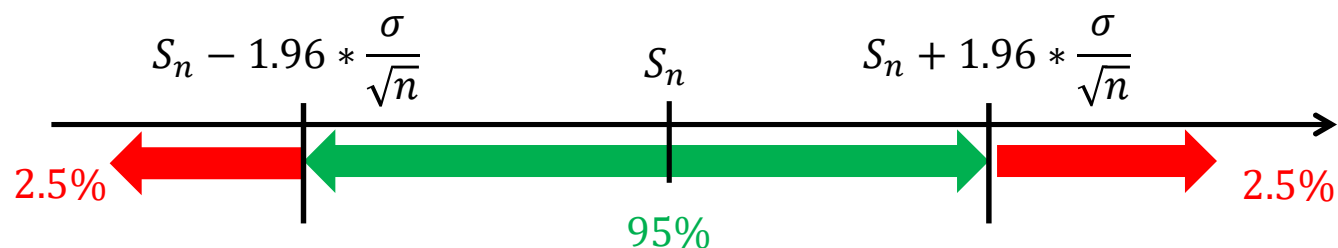
- Law of large numbers
 - Let $X_1, \dots, X_n, \dots$ be i.i.d. random variables with mean μ and variance σ^2
 - Define the sample mean as $S_n = \frac{1}{n} [X_1 + \dots + X_n]$
 - Then the sample mean converges to the real mean: $S_n \rightarrow \mu$ a.s.

- Central Limit Theorem

- The sample mean S_n
 - ✓ Is itself a random variable
 - ✓ It is normally distributed $N\left(\mu, \frac{\sigma^2}{n}\right)$



- We can now compute the confidence interval of μ as a function of S_n



Time series: Price

USD/MXN Historical Data



Time Frame:

Daily ▼



Download Data

14/03/2019 - 14/04/2019



Date :	Price :	Open :	High :	Low :	Change % :
Apr 12, 2019	18.7583	18.8335	18.8755	18.7494	-0.40%
Apr 11, 2019	18.8335	18.8185	18.9285	18.8124	0.09%
Apr 10, 2019	18.8165	18.9330	18.9500	18.8100	-0.61%
Apr 09, 2019	18.9312	18.9605	18.9838	18.8887	-0.19%
Apr 08, 2019	18.9675	19.0755	19.1367	18.9135	-0.56%
Apr 07, 2019	19.0734	19.0740	19.0900	19.0650	-0.00%
Apr 05, 2019	19.0738	19.1367	19.1635	19.0435	-0.38%
Apr 04, 2019	19.1474	19.2139	19.2822	19.1280	-0.38%
Apr 03, 2019	19.2200	19.2375	19.2600	19.1130	-0.04%
Apr 02, 2019	19.2280	19.1400	19.2610	19.0675	0.44%
Apr 01, 2019	19.1440	19.3770	19.3867	19.1360	-1.04%
Mar 31, 2019	19.3448	19.3803	19.3803	19.3424	-0.43%
Mar 29, 2019	19.4290	19.3435	19.4875	19.2705	0.45%
Mar 28, 2019	19.3425	19.3650	19.4505	19.3025	-0.10%
Mar 27, 2019	19.3625	19.1105	19.3875	19.1025	1.34%
Mar 26, 2019	19.1060	19.0150	19.1330	18.9930	0.46%
Mar 25, 2019	19.0188	19.1002	19.1602	18.9883	-0.53%
Mar 24, 2019	19.1206	19.1081	19.1245	19.0997	0.14%
Mar 22, 2019	19.0940	18.8600	19.1885	18.8435	1.23%
Mar 21, 2019	18.8621	18.8242	18.9305	18.7715	0.18%
Mar 20, 2019	18.8287	19.0143	19.0297	18.7425	-0.98%
Mar 19, 2019	19.0141	19.0505	19.0777	18.9795	-0.22%
Mar 18, 2019	19.0562	19.2013	19.2335	19.0416	-0.85%
Mar 17, 2019	19.2192	19.2025	19.2270	19.2025	0.06%
Mar 15, 2019	19.2071	19.3210	19.3259	19.1842	-0.58%
Mar 14, 2019	19.3185	19.2760	19.3773	19.2715	0.18%
Highest: 19.4875	Lowest: 18.7425	Difference: 0.7450	Average: 19.0968	Change %: -2.7226	

investing.com

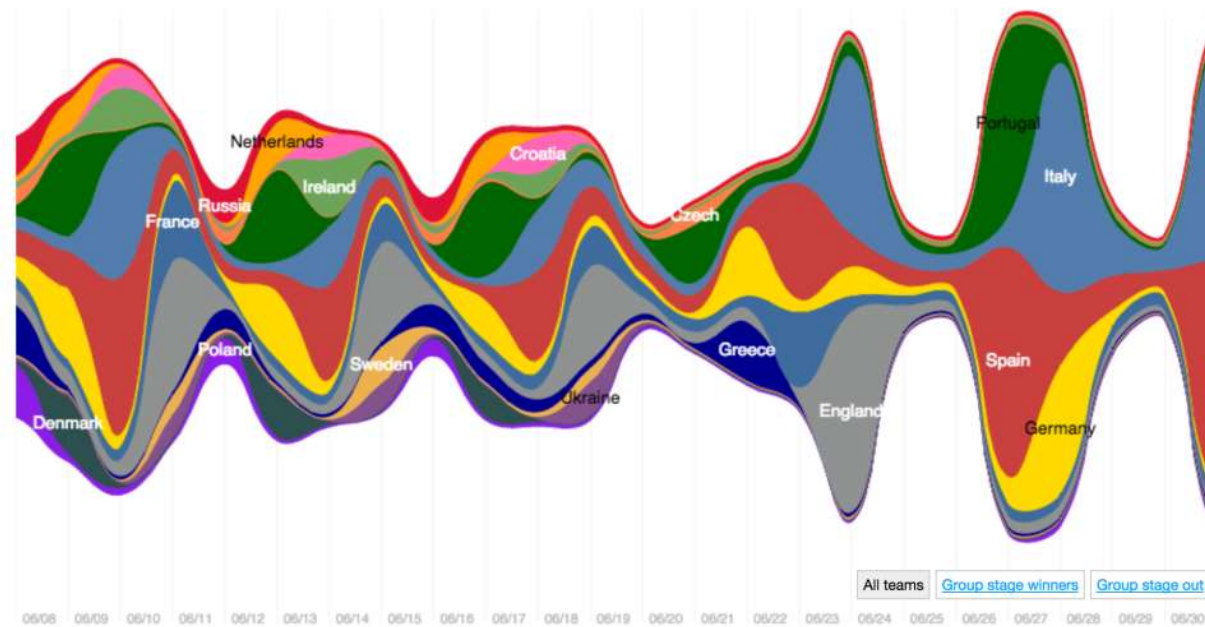
Time series: Volume



#Euro2012

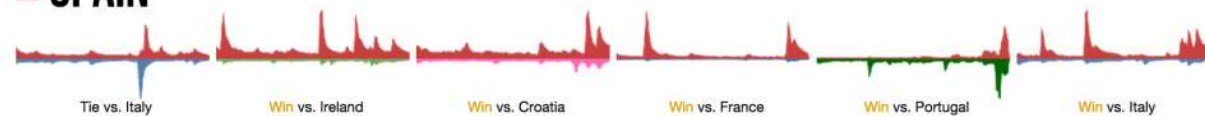
A summary for the action on Twitter during the European football tournament. [Tweet](#) 1,810

The Streamgraph below shows volume of Tweets during the #Euro2012 period. Click on a team's name to see details.

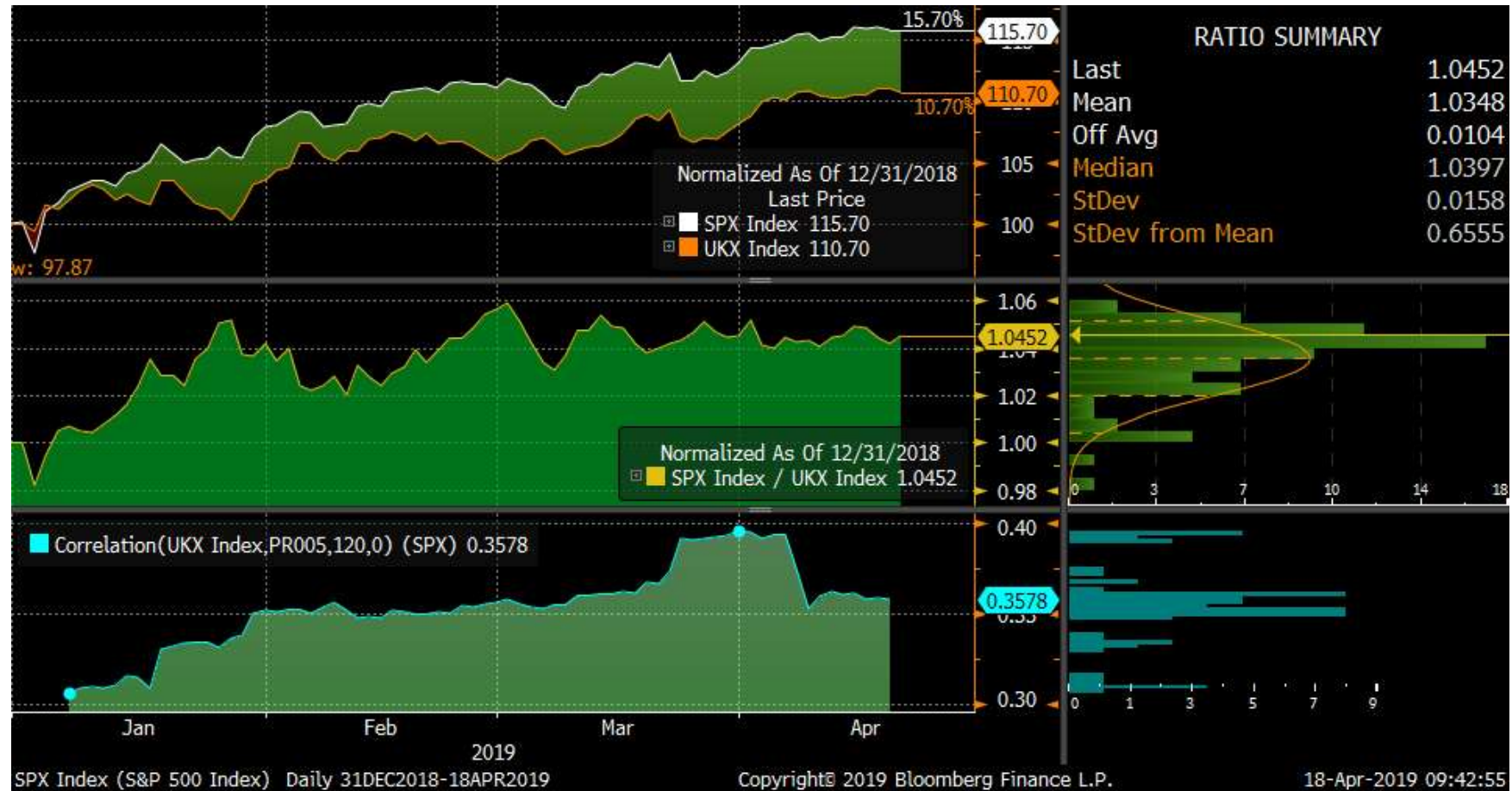


SPAIN

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Time series: Pairs trading



Comparing trading strategies

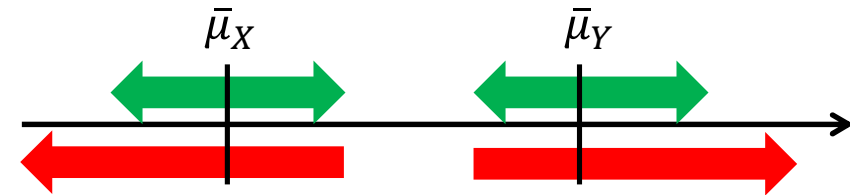
- We want to compare the performance of 2 trading strategies X and Y
 - At any time $t > 0$ we observe their performance, X_t and Y_t
- t is called the timestamp of the time series
 - It is the moment when the observations were taken
 - General format is 2019-04-26 11:55:47.235
- Generally we use the notation $t \in T$
 - T is an ordered subset on the real line
 - Discrete (finite) or interval (infinite)
- Notice that the index t is the same for both strategies X_t and Y_t
 - If not, we have asynchronous time series
 - There are full theories on how to deal with asynchronicity
 - This problem is very common in High Frequency Trading



Difference of performance

- We have a finite number of data points $t > 0$
 - X_t and Y_t where $t = 1, 2, \dots, N$

- Compute their sample means
 - $\bar{\mu}_X$ and $\bar{\mu}_Y$



- Suppose that we found that $\bar{\mu}_X < \bar{\mu}_Y$
 - Is it statistically significant?
- If we have disjoint confidence intervals
 - Problem solved
- If the confidence intervals intersect
 - Reduce the confidence level
 - Add more data points
 - Use a Student T-test for 2 means

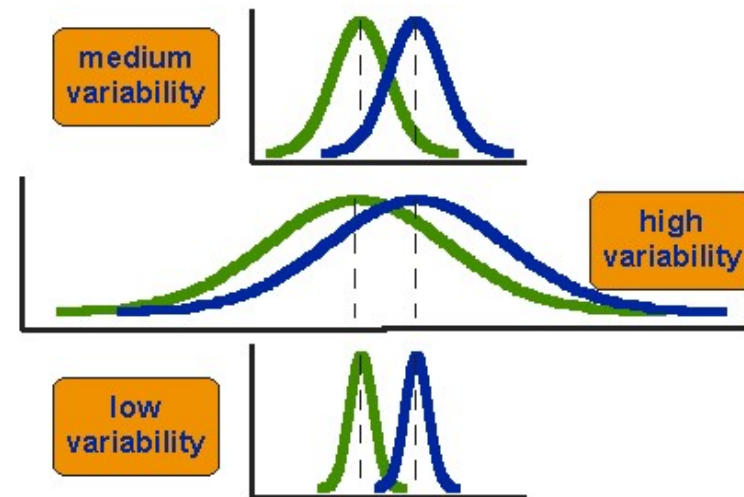
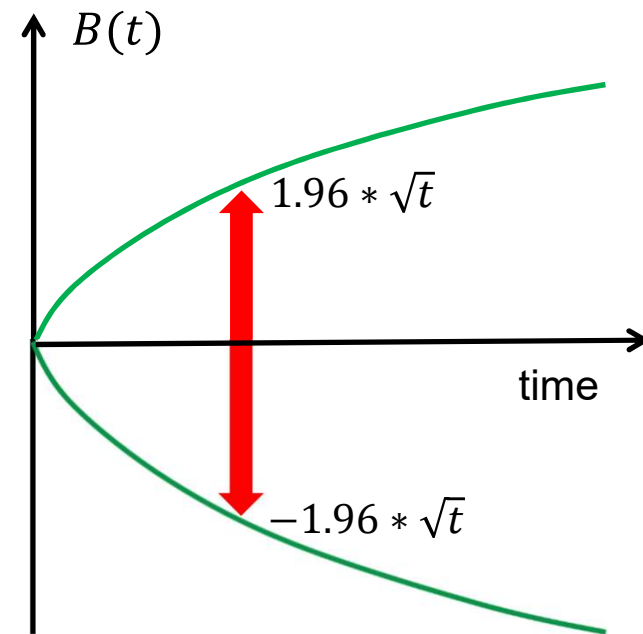


Figure 2. Three scenarios for differences between means.

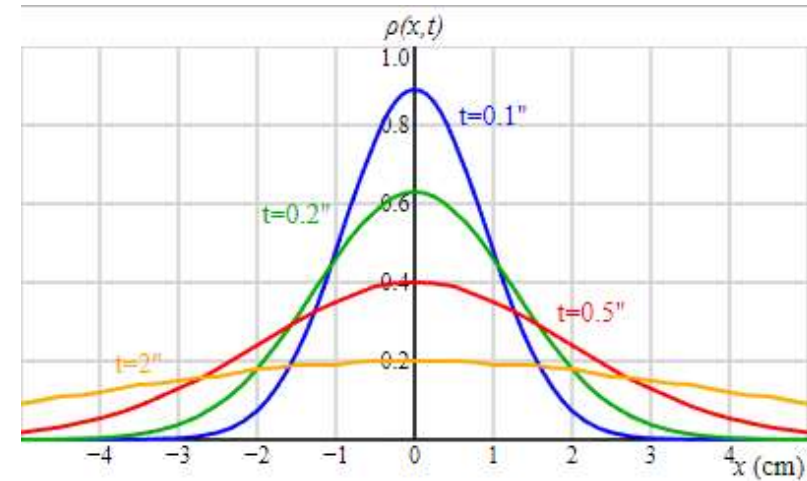
Brownian motion

- We can describe the Brownian motion based on
 - Normal distribution
 - Confidence intervals
- Start with a particle at the origin
 - $B(0) = 0$ a.s.
- At time t the particle moves randomly
 - We do not know the exact position
 - But we do know its distribution
 - ✓ $B(t) \sim N(0, t)$
- At time t the confidence interval 95% is
 - $[-1.96 * \sqrt{t}, +1.96 * \sqrt{t}]$



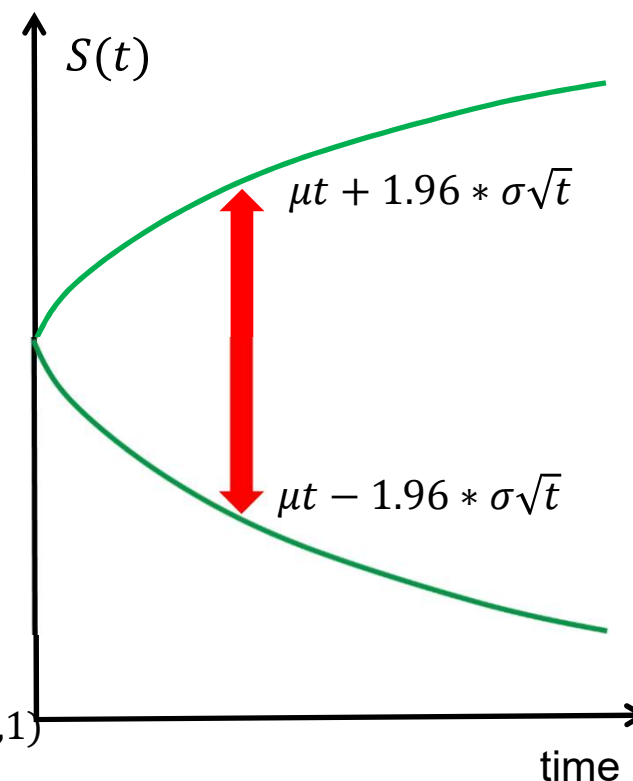
Properties of the Brownian motion

- Independence (Markov property)
 - $B(t + h) - B(t) \sim N(0, h)$
- Paths $t \rightarrow B(t)$
 - Are continuous
 - But not differentiable
- Scaling
 - If $B(t) \sim N(0, t)$ then
 - ✓ $S(t) = \mu + \sigma B(t) \sim N(\mu, \sigma^2 t)$
 - Confidence interval for $S(t)$ is
 - ✓ $[\mu - 1.96 * \sigma\sqrt{t}, \mu + 1.96 * \sigma\sqrt{t}]$



General Stochastic Processes

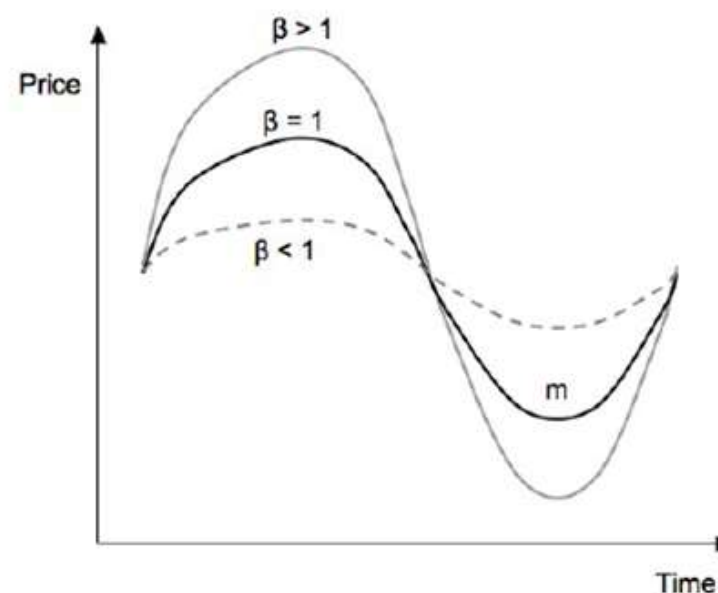
- Assume a standard Brownian motion
 - $B(t) \sim N(0, t)$
- We add drift and volatility
 - (1) $S(t) = \mu t + \sigma B(t)$
 - This is the classical model for financial returns
- Itô Process
 - Generalisation of the (1)
 - It is customary to use the differential notation
 - ✓ (2) $dS(t) = \mu(t, S)dt + \sigma(t, S)dB(t)$
- The discrete version of the Itô process (2) is
 - (3) $S(t + h) - S(t) = \mu(t, S)h + \sigma(t, S)B(h)$
 - Monte Carlo simulations assume (3) with $B(h) \sim \sqrt{h} N(0, 1)$
- Examples
 - Geometric Brownian motion: $dS(t) = \mu S(t)dt + \sigma S(t)dB(t)$
 - Ornstein-Uhlenbeck: $dS(t) = \theta(\mu - S(t))dt + \sigma dB(t)$



2. Geometry of the Capital Asset Pricing Model

Alphas and betas

- The Capital Asset Pricing Model is a linear regression of Asset a with respect to the market M
- A classical linear regression of the returns is
 - $r_a = \alpha + \beta r_M + \varepsilon$
 - $E[r_M] = E[\varepsilon] = 0$
- The error ε is called the **idiosyncratic risk**
 - It can be eliminated via diversification
 - ε is orthogonal to r_M i.e. $E[r_M \varepsilon] = 0$
- β is called the **systematic risk**
 - Exposure of r_a to the market
 - Cannot be diversified away
- α is called the **absolute return**
 - α is orthogonal to r_M and ε
 - It is the fraction of the total return r_a that cannot be explained by the market



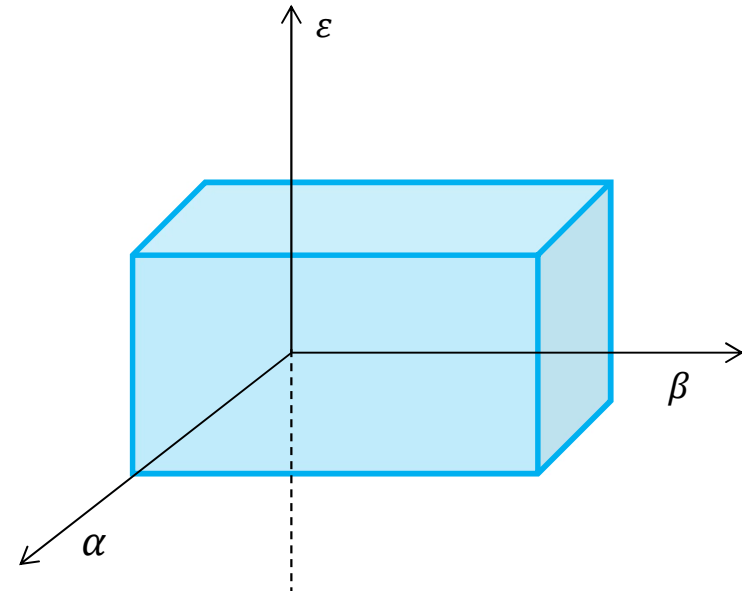
Efficient Market Theory

- We saw that CAPM means a linear model
 - $r_a = \alpha + \beta r_M + \varepsilon$
- Efficient Market Theory
 - ✓ No one can consistently beat the market
- Assuming $E[r_M] = 0$ and EMT
 - We necessarily have $\alpha = 0$
 - Otherwise we can construct a portfolio that always beats the market
 - ✓ $\alpha > 0$
 - ✓ $\beta = 0$
 - ✓ $\varepsilon \sim 0$



Dimension reduction

- We start with a **3D** problem
 - $r_a = \alpha + \beta r_M + \varepsilon$
- EMT implies reduction to **2D**
 - $r_a = \beta r_M + \varepsilon$
- Taking expectations get us to a **1D** problem
 - $E[r_a] = \beta E[r_M]$
- We can compute beta via covariance
 - $\beta = \frac{\text{Cov}(r_a, r_M)}{\text{Var}(r_M)} = \rho(r_a, r_M) \frac{\sigma_A}{\sigma_M}$
- Therefore, **beta is a volatility-adjusted correlation**



Variance and covariance

- As usual, we start with
 - (*) $r_a = \beta r_M + \varepsilon$
- Squaring (*) and taking expectations we obtain
 - $Var(r_a) = \beta^2 Var(r_M) + Var(\varepsilon)$
- Therefore, we can approximate $Var(r_a) \sim \beta^2 Var(r_M)$
 - But this underestimates $Var(r_a)$
 - Esp. for big errors
- Similarly, for the covariance of 2 assets
 - $Cov(r_a, r_b) = \beta_a \beta_b Var(r_M) + Cov(\varepsilon_a, \varepsilon_b)$
- Therefore, we can approximate $Cov(r_a, r_b) \sim \beta_a \beta_b Var(r_M)$
 - But this assumes $Cov(\varepsilon_a, \varepsilon_b) = 0$
 - Which is a BIG assumption



Classification of investment strategies

- **Index tracker:**

- Replicate the performance of a benchmark
- It can be an index (S&P 500, BMV, Stoxx600), a commodity, etc
- $\beta = 1, \alpha = 0$

- **Traditional long-only asset manager:**

- Outperform the market with an extra, uncorrelated return
- $\beta = 1, \alpha > 0$



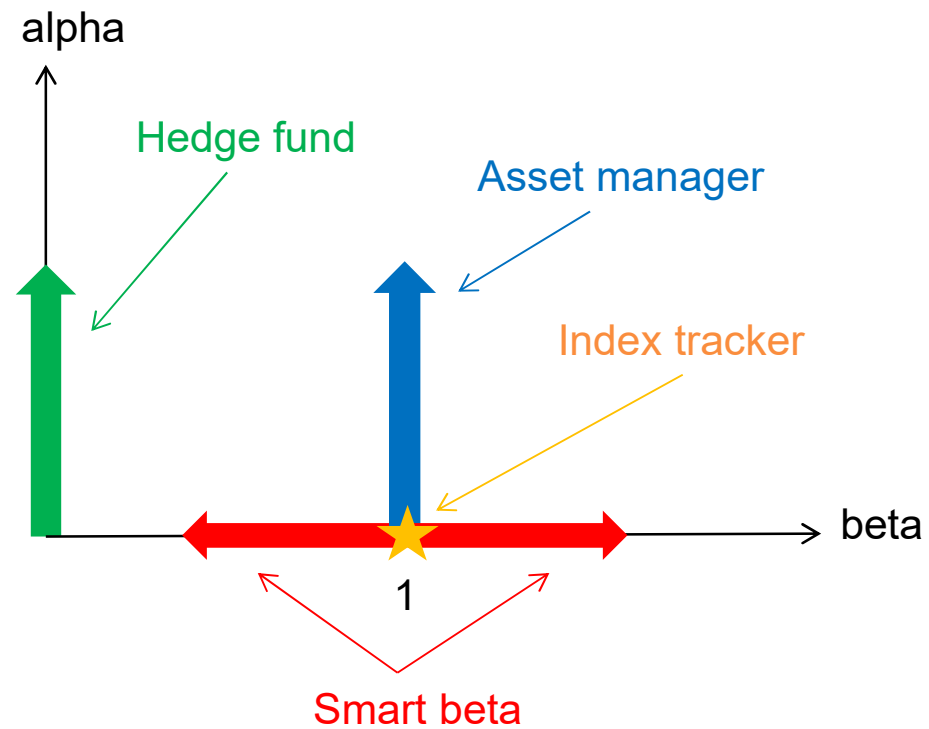
- **Smart beta:**

- Outperform the market by dynamically adjusting your portfolio weights
- $\beta > 1$ when the market is up
- $\beta < 1$ when the market is down
- $\alpha = 0$

- **Hedge fund:**

- Deliver absolute returns that are not correlated with the market
- $\beta = 0, \alpha > 0$

Geometric classification of funds



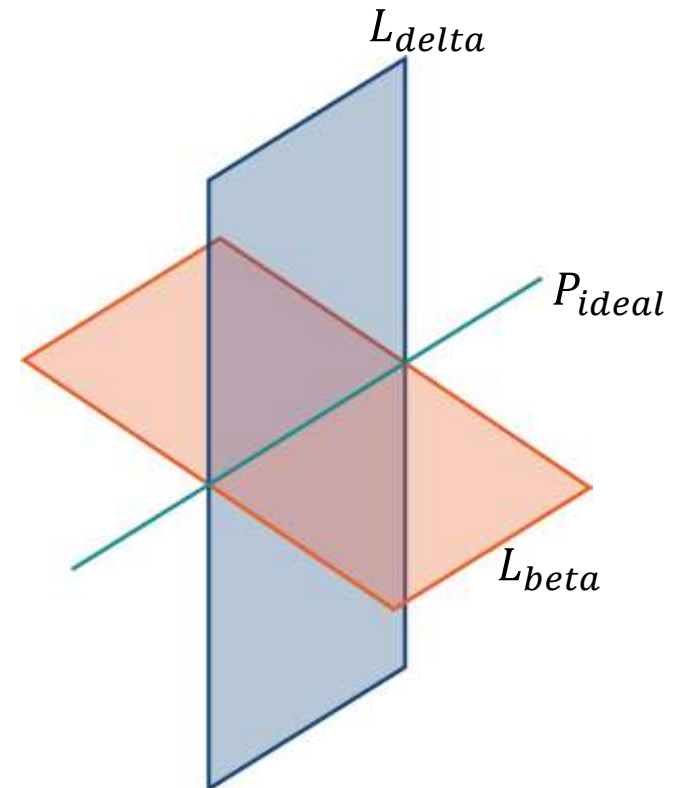
Delta neutral vs beta neutral

- We start with a given security (can be a portfolio)
 - Let S_0 be its value in USD
- Hedging with one security
 - Delta Neutral
 - ✓ Find S_1 such that $S_0 + S_1 = 0$
 - Beta Neutral
 - ✓ Find S_1 such that $\beta_0 S_0 + \beta_1 S_1 = 0$
- Hedging with N securities
 - Delta Neutral
 - ✓ Find $S_1, \dots, S_N$ such that $S_0 + \sum S_n = 0$
 - Beta Neutral
 - ✓ Find $S_1, \dots, S_N$ such that $\beta_0 S_0 + \sum \beta_n S_n = 0$



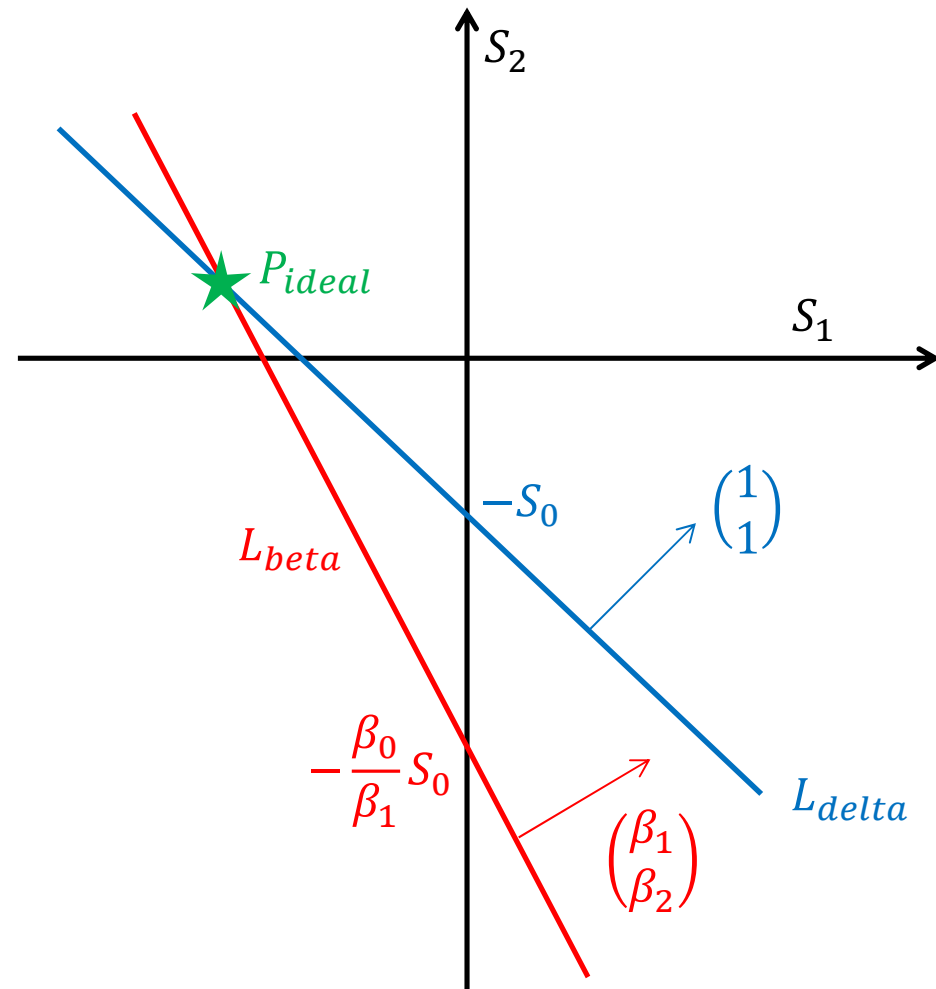
Beta and delta neutral hyperplanes

- Delta-neutral hyperplane in R^N
 - $L_{delta} = \{S_0 + \sum S_n = 0\}$
 - All points over L_{delta} give a delta-neutral solution
- Beta-neutral hyperplane in R^N
 - $L_{beta} = \{\beta_0 S_0 + \sum \beta_n S_n = 0\}$
 - All points over L_{beta} give a beta-neutral solution
- Ideal solution:
 - $P_{ideal} = L_{delta} \cap L_{beta}$
 - P_{ideal} is a hyperplane of dimension $N - 2$
 - All points over P_{ideal} are beta and delta neutral



Beta and delta neutral lines

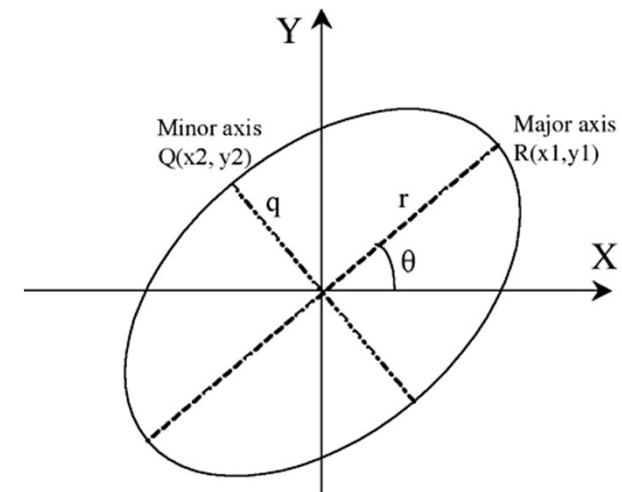
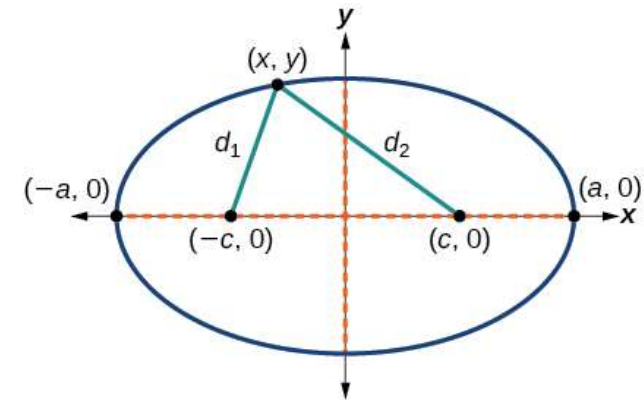
- Suppose $\beta_0 > \beta_1 > \beta_2$
- $L_{delta} = \{S_0 + S_1 + S_2 = 0\}$
- $L_{beta} = \{\beta_0 S_0 + \beta_1 S_1 + \beta_2 S_2 = 0\}$
- $P_{ideal} = L_{delta} \cap L_{beta}$
 - $S_1 < 0$
 - $S_2 > 0$



3. Geometry of the variance-covariance matrix

Review of Analytical Geometry

- General form of an ellipse
 - (1) $Ax^2 + 2Bxy + Cy^2 + Dx + Ey + F = 0$
- Change of variables
 - With a rotation and a translation
 - We can eliminate the linear and cross terms
 - And reduce the equation (1) to
 - ✓ (2) $Ax^2 + By^2 + C = 0$
- Refactoring (2) we obtain
 - (3) $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$
- In matrix notation, (3) becomes
 - (4) $(x, y) \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 1$ where $\lambda_1 = \frac{1}{a^2}$ and $\lambda_2 = \frac{1}{b^2}$

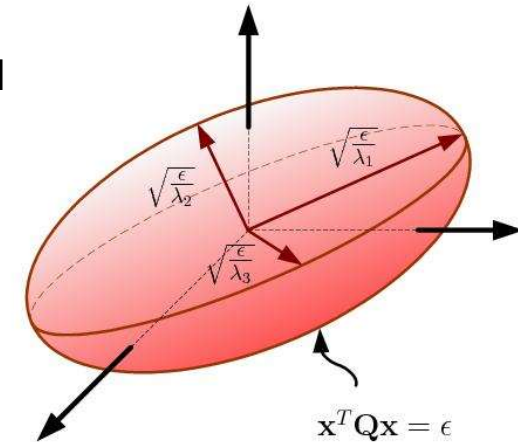


Eigenvalues and eigenvectors

- In summary, we need a change of coordinates
 - Such that under the new coordinates the matrix is diagonal

$$\begin{bmatrix} A & B \\ B & C \end{bmatrix} \rightarrow \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

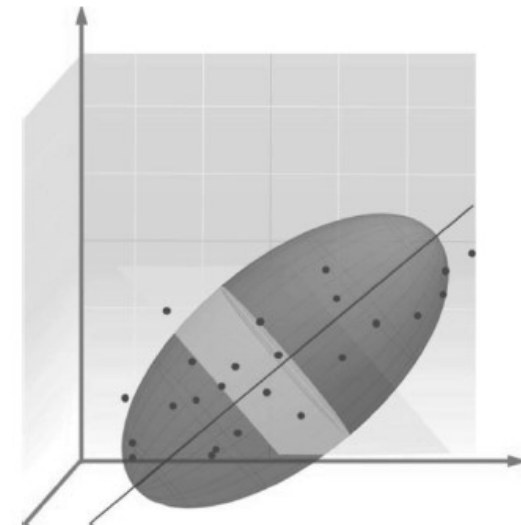
- What matrices accept diagonalisation?
 - Symmetric: $a_{ij} = a_{ji}$
 - Positive semi-definite: $x^T Q x \geq 0$ for any $x \in R^N$



- A bit of notation in N dimensions

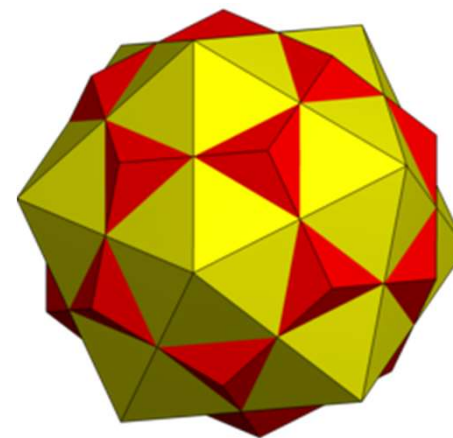
$$Q \rightarrow \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_N \end{bmatrix} \text{ is the change of variables}$$

- λ_n is called an eigenvalue
- The vector associated to λ_n is called eigenvector
 - ✓ $Q v_n = \lambda_n v_n$



Variance-covariance matrix

- Let $X_1, \dots, X_N$ be random variables with means $\mu_1, \dots, \mu_N$ resp.
 - The variance-covariance matrix Q is such that
 - $Q_{ij} = \text{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$
- By definition, Q is **symmetric**
 - $Q_{ij} = Q_{ji}$
- Q is always **positive semi-definite**
 - Define $X = \begin{pmatrix} X_1 \\ \vdots \\ X_N \end{pmatrix}$ and $\mu = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_N \end{pmatrix}$
 - Then $Q(X) = E[(X - \mu)(X - \mu)^T]$
 - For any $w \in R^N$ we have
 - ✓ $w^T Q(X) w = E[w^T (X - \mu)(X - \mu)^T w] = E\left[\left((X - \mu)^T w\right)^2\right] \geq 0$

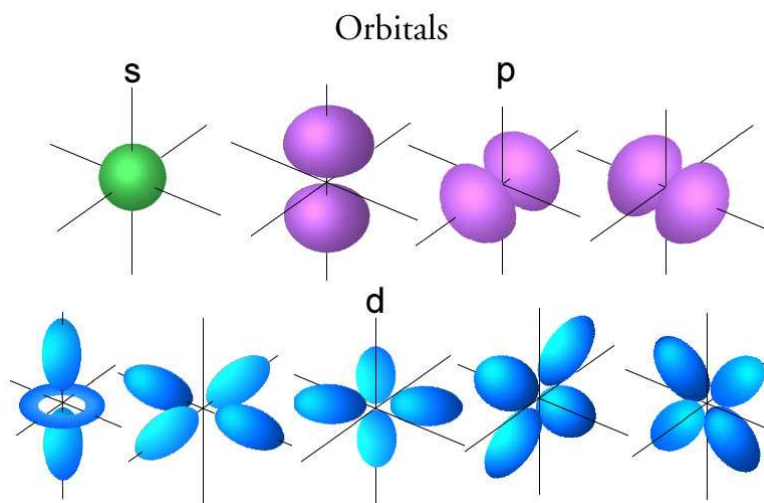


Orthogonality of eigenvectors

- Let Q be a symmetric matrix
- Let v_i and v_j be two eigenvectors with $\lambda_i \neq \lambda_j$
- If $\lambda_i \neq 0$ and $\lambda_j \neq 0$ then:

$$\begin{aligned}\lambda_i(v_i^T v_j) &= (\lambda_i v_i^T) v_j = (\lambda_i v_i)^T v_j \\ &= (Q v_i)^T v_j = v_i^T Q^T v_j = v_i^T Q v_j \\ &= v_i^T (Q v_j) = v_i^T (\lambda_j v_j) = \lambda_j (v_i^T v_j)\end{aligned}$$

- Since $\lambda_i \neq \lambda_j$ then necessarily $v_i^T v_j = 0$



How to use the covariance?

- The variance-covariance matrix Q is symmetric and positive semi-definite

➤ There is a change of variables such that

$$✓ Q \rightarrow \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_N \end{bmatrix} \text{ where } 0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$$

- The smallest eigenvalue λ_1 of Q

➤ Has an eigenvector v_1 that minimises the variance
 ➤ This v_1 is called the **minimum variance portfolio**

- The largest eigenvalue λ_N of Q

➤ Has an eigenvector v_N that maximises the variance
 ➤ This v_N is called the **maximum variance portfolio or the PCA portfolio**

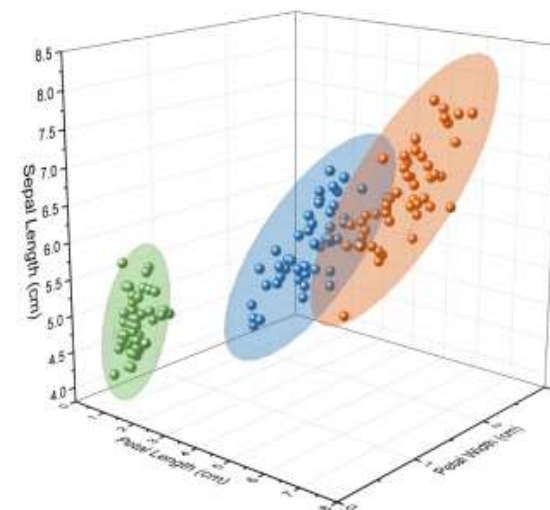
- The eigenvalues provide lower and upper bounds for the volatility of a portfolio:

➤ For any portfolio $w = (w_1, \dots, w_N)$ in $\mathbb{R}^N$ such that $\|w\| = 1$ we have

$$\lambda_1 \leq w^T Q w \leq \lambda_N$$

➤ Since $w^T Q w$ is the variance of the portfolio then its volatility σ_w satisfies

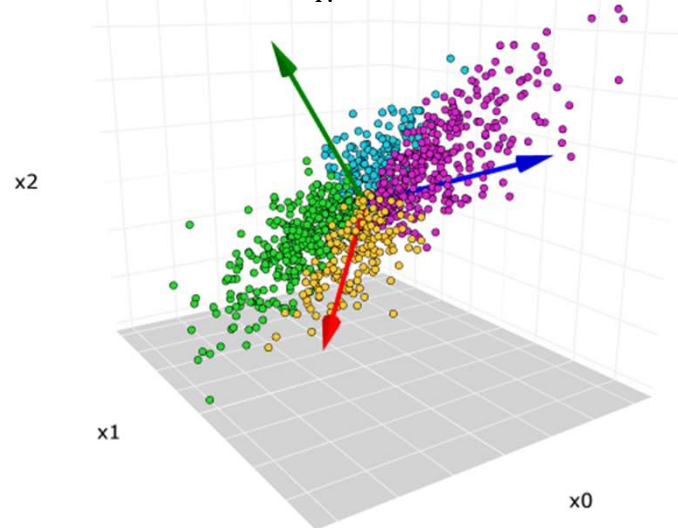
$$\sqrt{\lambda_1} \leq \sigma_w \leq \sqrt{\lambda_N}$$



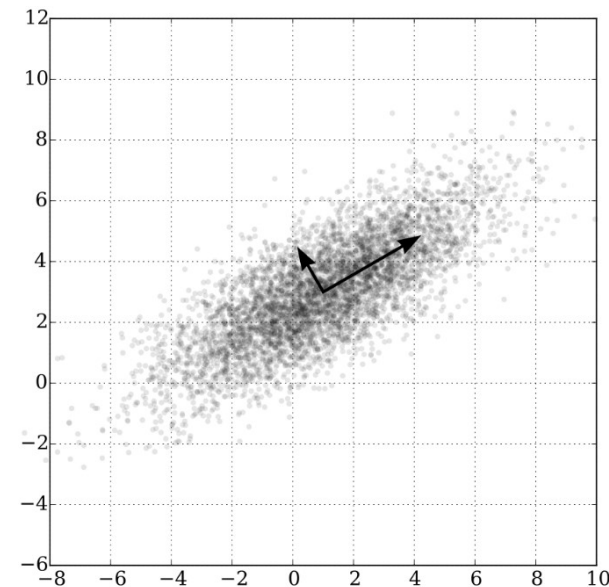
PCA: Principal Component Analysis

- Goals of the PCA
 - Dimension reduction: in general for visualisation
 - The reduced dimensions describe MOST of the variation in the data
 - In general, 2 dimensions are used for plotting the data
 - Lots of applications: Finance, Marketing, Business Intelligence, etc
- We will use the spectral properties of the variance-covariance matrix:

$$Q \rightarrow \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_N \end{bmatrix}$$



https://en.wikipedia.org/wiki/Principal_component_analysis

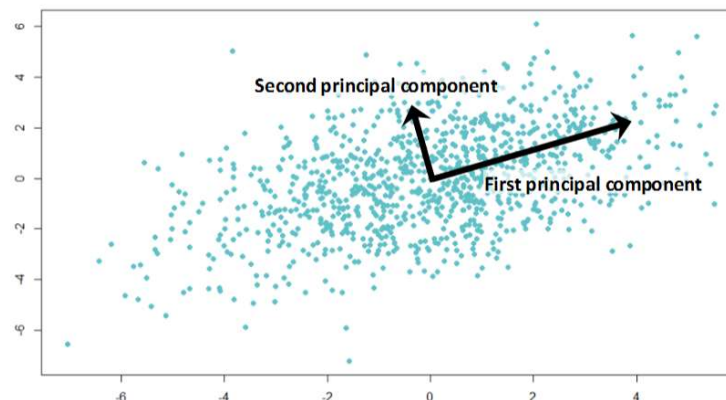


PCA recipe

- Pick a threshold:
 - Say that we want to keep at least 80% of the total variance
- Pick the eigenvector v_N for the largest largest eigenvalue λ_N
- Compute the “variance” explained by v_N :

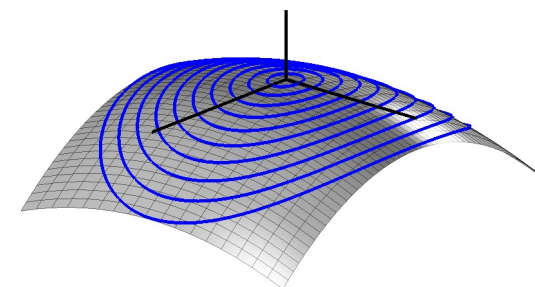
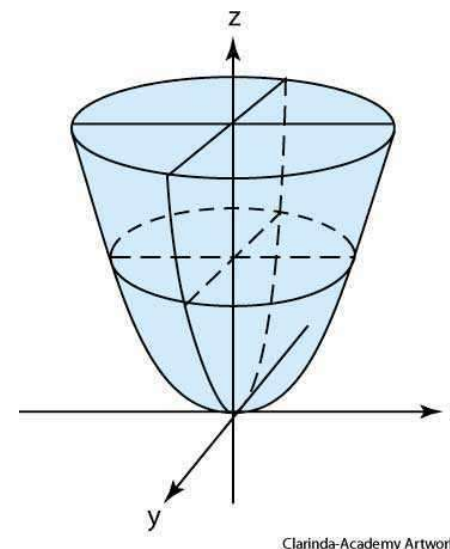
$$VAR(v_N) = \frac{\lambda_N}{\sum \lambda_i}$$

- If $VAR(v_N) < 0.8$ then add the second eigenvector v_{N-1} for the second largest eigenvalue λ_{N-1}
 - Compute the “variance” explained by (v_{N-1}, v_N) :
- $$VAR(v_{N-1}, v_N) = \frac{\lambda_{N-1} + \lambda_N}{\sum \lambda_i}$$
- If $VAR(v_{N-1}, v_N) < 0.8$ then add the next eigenvector for the next largest eigenvalue
 - Stop at the first $K < N$ eigenvectors such that $VAR(v_{N-K+1}, \dots, v_N) \geq 0.8$



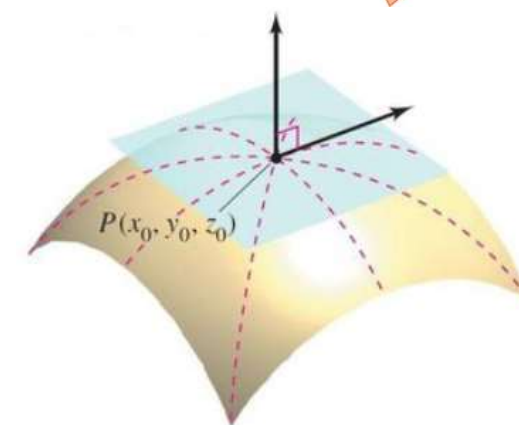
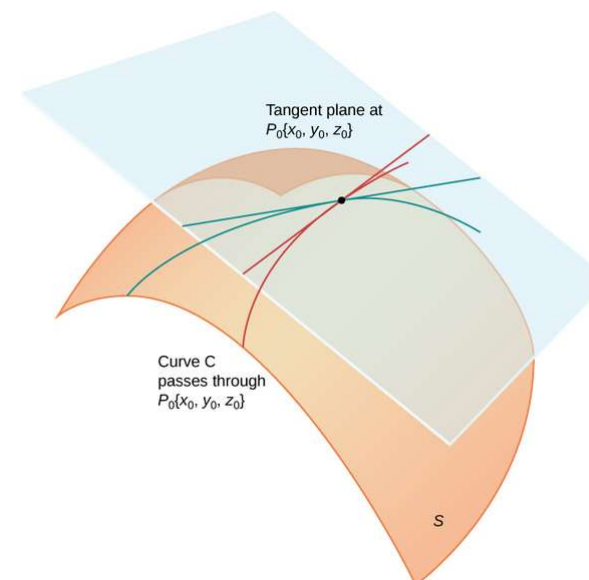
Level curves and risk reduction

- We start with the variance of a portfolio
 - $z = f(w) = w^T Q w$, $w \in \mathbb{R}^N$
 - It corresponds to an N dimensional paraboloid
- All points on the level curve have the same height
 - Same level of risk in all portfolios on the curve
 - Risk-level curves
 - I call them **iso-risk curves**
- Risk reduction
 - Move from a higher level curve to a lower one
 - The best direction is the one that reduces the risk the fastest
 - This means descending following the path of $-\nabla f(x)$
 - This is known as **gradient descent**
- Why this direction?
 - Define $t \rightarrow F(t) := f(x(t))$, $x(0) = p$
 - Best direction $x(t)$ is where $F'(0)$ is the most negative
 - But $F'(0) = \nabla f(p)^T x'(0) = \|\nabla f(p)\| \|x'(0)\| \cos \theta$
 - Hence $x'(0) = -\nabla f(p)$



Optimisation with constraints

- We start with the variance of a portfolio, again
 - $f(w) = w^T Q w$, $w \in \mathbb{R}^N$
 - The objective is to minimise the variance $f(w)$
- Without constraints
 - w^u is the eigenvector of smallest eigenvalue
 - Minimum variance portfolio
- With constraints of the form $c(x) \geq 0$
 - Positive weights: $w_n \geq 0$
 - Target return: $\sum w_n r_n = r_p$
 - This problem is known as the **Markowitz Portfolio**
- The minimum w^c satisfies $\nabla f(w^c) + \lambda \nabla c(w^c) = 0$
 - Define $t \rightarrow F(t) := f(x(t))$, $x(0) = p$
 - If p is a minimum then $F'(0) = \nabla f(p)^T x'(0) = 0$
 - If $x(t)$ is on the surface $\{c(x) = 0\}$ then $c(x(t)) \equiv 0$
 - Therefore $\nabla c(p)^T x'(0) = 0$
 - In consequence, $\nabla f(w^c)$ and $\nabla c(w^c)$ are collinear

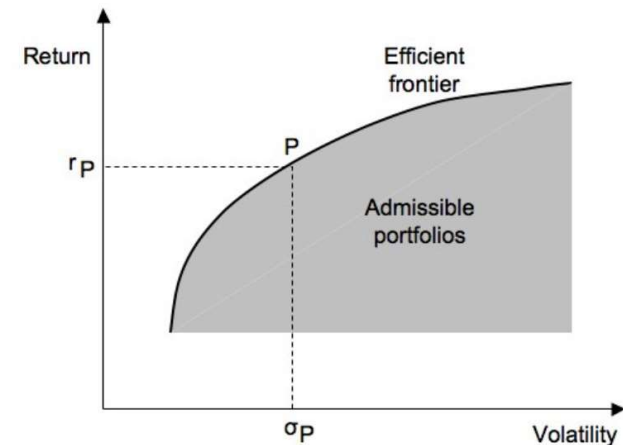


Tangent plane to surface S at P

Efficient frontier

The Markowitz Portfolio can be understood in the plane risk-return

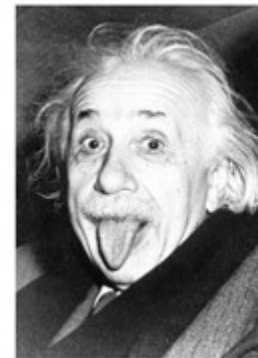
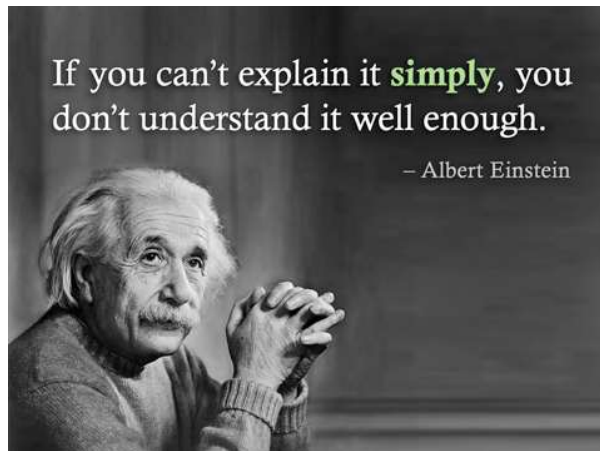
- Admissible portfolios are those that satisfy the constraints
 - Positive weights
 - Target portfolio return
 - Target portfolio risk
- There are two interpretations of an optimal portfolio
 - Given a target level of return r_p , minimise the risk
 - Given a target level of risk σ_p , maximise the return
- Efficient frontier
 - Separates admissible and non-admissible portfolios
 - Portfolios on the Efficient Frontier are optimal



4. Conclusions

In a nutshell

- Geometry of confidence intervals
 - It is all about expanding and contracting with volatility (and time)
- Geometry of CAPM
 - It is all about linear regressions and intersections of hyperplanes
- Geometry of the variance-covariance matrix
 - It is all about eigenvalues, eigenvectors and level curves
- Slides available online at <https://fractalvelvet.wordpress.com/>



"Everything should be made as simple as possible. But not simpler."

-Albert Einstein

Thank you for your
attention

